

Quiz — 1.3.4 Web Technologies

Name: _____ Date: _ **Mark:** ____ / 25

OCR H446 · Component 01 · 1.3.4

Section A — Multiple choice (8 marks)

1. The role of HTML in a web page is to define the: A. Colour and font styling B. Structure and content C. Server-side database logic D. Encryption of the page

Your answer: ☐

2. Which technology is primarily responsible for the presentation/styling of a page? A. HTML B. CSS C. JavaScript D. SQL

Your answer: ☐

3. A CSS selector beginning with # targets an element by its: A. tag name B. class C. id D. attribute value

Your answer: ☐

4. JavaScript is most commonly used to provide: A. Page structure B. Client-side interactivity and behaviour C. Database storage D. IP addressing

Your answer: ☐

5. In PageRank, a link from one page to another is treated as: A. A penalty B. A "vote" for the linked page's importance C. A keyword in the index D. An encryption key

Your answer: ☐

6. Links from high-ranking pages in PageRank are: A. Ignored B. Worth more than links from low-ranking pages C. Always worth the same as any other link D. Counted as negative votes

Your answer: ☐

7. Which is an example of server-side processing? A. Instant form validation in the browser B. Animating a button on hover C. Checking login credentials against a database D. Showing/hiding a menu on click

Your answer: ☐

8. Why must security-critical validation also be performed server-side? A. It runs faster on the client B. Client-side checks can be viewed and bypassed by the user C. The server cannot run JavaScript D. It reduces server load

Your answer: ☐

Section B — Short answer (12 marks)

9. State the role of each of HTML, CSS and JavaScript in a web page. [3]

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10. Describe how a search engine builds its index of web pages. [2]

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11. Explain the purpose of the damping factor in the PageRank algorithm. [2]

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12. PageRank is calculated iteratively. Explain what this means and when it stops. [2]

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13. Give one advantage and one disadvantage of using client-side processing instead of server-side processing. [2]

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14. State one task that is better suited to server-side processing and explain why. [1]

Section C — Exam-style question (5 marks)

15. An online shop has a checkout form. When a customer submits the form, the email address format is checked instantly in the browser, and the payment details are then verified against the bank's records.

Explain why the email check can be done client-side but the payment verification must be done server-side, referring to speed, server load and security. [5]

This image shows a blank sheet of white paper with horizontal ruling lines. The top section features five solid black lines spaced evenly apart. Below these, there are four more solid black lines, also spaced evenly. At the bottom of the page, there are ten dashed blue lines, providing a different style of guidance for writing. The entire sheet is otherwise empty, with no text or other markings.